

Afwan Apriansyah

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Education

Universitas Gunadarma

Bachelor of Science in Informatics (Current GPA: 3.62/4.00)

Expected September 2026

Depok, East Java

Undergraduate Thesis

powerbind — Home Power Usage Management System with Integrated Multimodal AI Agent (Groq)

In Progress

- Designed and implemented a multi-room IoT system using ESP32 microcontrollers and LD2410C mmWave radar sensors for real-time human presence detection, with automated relay-based actuation.
- Built a Spring Boot backend with MQTT integration via Mosquitto broker, WebSocket push via STOMP, time-series data storage in InfluxDB, relational data in PostgreSQL, and Redis-backed rate limiting with login-lockout security.
- Developed a Vue.js real-time web dashboard displaying live room occupancy, relay status, power consumption (PZEM-004T), estimated PLN electricity cost, and historical usage charts with WebSocket-driven updates.
- Integrated a multimodal AI agent powered by Groq API (LLaMA) and Whisper for voice-to-text, enabling natural-language queries over energy consumption data retrieved from InfluxDB.

Projects

AI Agent Chat Assistant | *Vue.js, Java, Spring Boot, PostgreSQL, Redis, Docker*

- Built a fullstack AI chat application with an agentic assistant capable of native tool-calling (web search, URL reading) and streaming responses in real time via Server-Sent Events.
- Designed a secure authentication system with JWT access tokens and rotating refresh tokens for silent session renewal.
- Managed the project backlog and sprint workflow in Jira, tracking features and bug fixes through an Agile board.

PT. Pandawa Digital Mandiri Website | *TypeScript, Angular, Java, Spring Boot, PostgreSQL, Redis, Docker*

- Developed and enhanced the company's website as a fullstack developer, adding admin functionalities and TPS features.
- Implemented a lead scoring system powered by Gemini AI, integrating it with REST APIs for seamless data flow.
- Built a robust dashboard with CRUD operations, improving internal workflow efficiency and system maintainability.

Car Rental Management System | *Java, Spring Boot, NetBeans*

- Developed a Java-based desktop application to manage car rental operations, including car availability, customer data, and rental transactions.
- Implemented object-oriented design with clear separation between GUI components and data models to improve maintainability.
- Built interactive Swing interfaces for managing cars, customers, and rental status within a single unified application.

Technical Skills

Technologies: Spring Boot, Vue.js, Angular, PostgreSQL, Redis, Docker

IoT & Embedded Systems: ESP32, MQTT (Mosquitto), LD2410C mmWave, PZEM-004T, InfluxDB, Relay-based actuation